MSG No. 224.114 $Pn'\bar{3}'m'$ [Type III, cubic]

Table 1: Wyckoff site: 2a, site symmetry: -4'3m'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,8,9,10,17,18,19,20,21,22,23,24,26,27,28,29,30,31,35,36,37,38,39,40]
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[2,3,4,5,6,7,11,12,13,14,15,16,25,32,33,34,41,42,43,44,45,46,47,48]

Table 2: Wyckoff site: 4b, site symmetry: .-3'm'

No.	position	mapping
1	$[0, 0, 0]$	[1,12,14,16,17,18,25,36,38,40,41,42]
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[2,7,9,15,19,24,26,31,33,39,43,48]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[3,4,10,11,20,21,27,28,34,35,44,45]
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[5,6,8,13,22,23,29,30,32,37,46,47]

Table 3: Wyckoff site: 4c, site symmetry: .-3'm'

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[1,12,14,16,17,18,25,36,38,40,41,42]
2	$\left[0, \frac{1}{2}, 0\right]$	[2,7,9,15,19,24,26,31,33,39,43,48]
3	$\left[0, 0, \frac{1}{2}\right]$	[3,4,10,11,20,21,27,28,34,35,44,45]
4	$\left[\frac{1}{2}, 0, 0\right]$	[5,6,8,13,22,23,29,30,32,37,46,47]

Table 4: Wyckoff site: 6d, site symmetry: -4'2.m'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[1,8,9,10,26,27,37,38]
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[2,3,13,14,25,32,33,34]
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[4,5,15,16,42,43,45,47]
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[6,7,11,12,41,44,46,48]
5	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[17,20,22,24,30,31,35,36]
6	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[18,19,21,23,28,29,39,40]

Table 5: Wyckoff site: 8e, site symmetry: .3m'

No.	position	mapping
1	$[x, x, x]$	[1,17,18,36,38,40]
2	$\left[x+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	[2,7,15,33,43,48]
3	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	[3,4,11,34,44,45]
4	$\left[-x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	[5,6,13,32,46,47]
5	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	[8,22,23,29,30,37]
6	$\left[\frac{1}{2}-x, x, \frac{1}{2}-x\right]$	[9,19,24,26,31,39]
7	$\left[\frac{1}{2}-x, \frac{1}{2}-x, x\right]$	[10,20,21,27,28,35]

8	$[-x, -x, -x]$	[12,14,16,25,41,42]

Table 6: Wyckoff site: 12f, site symmetry: 2.22

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[1,8,13,14]
2	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	[2,3,9,10]
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[4,16,18,21]
4	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[5,15,19,23]
5	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	[6,11,20,22]
6	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	[7,12,17,24]
7	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[25,32,37,38]
8	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	[26,27,33,34]
9	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[28,40,42,45]
10	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[29,39,43,47]
11	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	[30,35,44,46]
12	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	[31,36,41,48]

Table 7: Wyckoff site: 12g, site symmetry: 2.m'm'

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	[1,8,37,38]
2	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	[2,3,33,34]
3	$\left[\frac{3}{4}, \frac{3}{4}, -x\right]$	[4,16,42,45]
4	$\left[\frac{3}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[5,15,43,47]
5	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[6,11,44,46]
6	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	[7,12,41,48]
7	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[9,10,26,27]
8	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	[13,14,25,32]
9	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[17,24,31,36]
10	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[18,21,28,40]
11	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[19,23,29,39]
12	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[20,22,30,35]

Table 8: Wyckoff site: 24h, site symmetry: 2..

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	[1,8]
2	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[2,3]
3	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[4,16]
4	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[5,15]
5	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[6,11]
6	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	[7,12]
7	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	[9,10]
8	$\left[-x, \frac{1}{4}, \frac{3}{4}\right]$	[13,14]

continued ...

Table 8

No.	position	mapping
9	$\left[\frac{3}{4}, x, \frac{1}{4}\right]$	[17,24]
10	$\left[\frac{1}{4}, \frac{3}{4}, x\right]$	[18,21]
11	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	[19,23]
12	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[20,22]
13	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	[25,32]
14	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{1}{4}\right]$	[26,27]
15	$\left[\frac{3}{4}, \frac{1}{4}, x\right]$	[28,40]
16	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[29,39]
17	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[30,35]
18	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	[31,36]
19	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[33,34]
20	$\left[x, \frac{3}{4}, \frac{1}{4}\right]$	[37,38]
21	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[41,48]
22	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[42,45]
23	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	[43,47]
24	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[44,46]

Table 9: Wyckoff site: 24i, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{2}, y, y+\frac{1}{2}\right]$	[1,13]
2	$\left[0, \frac{1}{2}-y, y+\frac{1}{2}\right]$	[2,10]
3	$[0, y, -y]$	[3,9]
4	$\left[y, y+\frac{1}{2}, \frac{1}{2}\right]$	[4,18]
5	$\left[\frac{1}{2}-y, y+\frac{1}{2}, 0\right]$	[5,19]
6	$[-y, 0, y]$	[6,20]
7	$\left[y+\frac{1}{2}, \frac{1}{2}, y\right]$	[7,17]
8	$\left[\frac{1}{2}, \frac{1}{2}-y, -y\right]$	[8,14]
9	$\left[y+\frac{1}{2}, 0, \frac{1}{2}-y\right]$	[11,22]
10	$\left[-y, \frac{1}{2}, \frac{1}{2}-y\right]$	[12,24]
11	$[y, -y, 0]$	[15,23]
12	$\left[\frac{1}{2}-y, -y, \frac{1}{2}\right]$	[16,21]
13	$\left[\frac{1}{2}, -y, \frac{1}{2}-y\right]$	[25,37]
14	$\left[0, y+\frac{1}{2}, \frac{1}{2}-y\right]$	[26,34]
15	$[0, -y, y]$	[27,33]
16	$\left[-y, \frac{1}{2}-y, \frac{1}{2}\right]$	[28,42]
17	$\left[y+\frac{1}{2}, \frac{1}{2}-y, 0\right]$	[29,43]
18	$[y, 0, -y]$	[30,44]
19	$\left[\frac{1}{2}-y, \frac{1}{2}, -y\right]$	[31,41]
20	$\left[\frac{1}{2}, y+\frac{1}{2}, y\right]$	[32,38]
21	$\left[\frac{1}{2}-y, 0, y+\frac{1}{2}\right]$	[35,46]
22	$\left[y, \frac{1}{2}, y+\frac{1}{2}\right]$	[36,48]
23	$[-y, y, 0]$	[39,47]
24	$\left[y+\frac{1}{2}, y, \frac{1}{2}\right]$	[40,45]

Table 10: Wyckoff site: `24j`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{1}{2}, y, -y\right]$	`[1,14]`
2	$\left[0, y, y+\frac{1}{2}\right]$	`[2,9]`
3	$\left[0, \frac{1}{2}-y, -y\right]$	`[3,10]`
4	$\left[\frac{1}{2}-y, y+\frac{1}{2}, \frac{1}{2}\right]$	`[4,21]`
5	$\left[y, y+\frac{1}{2}, 0\right]$	`[5,23]`
6	$\left[-y, 0, \frac{1}{2}-y\right]$	`[6,22]`
7	$\left[y+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-y\right]$	`[7,24]`
8	$\left[\frac{1}{2}, \frac{1}{2}-y, y+\frac{1}{2}\right]$	`[8,13]`
9	$\left[y+\frac{1}{2}, 0, y\right]$	`[11,20]`
10	$\left[-y, \frac{1}{2}, y\right]$	`[12,17]`
11	$\left[\frac{1}{2}-y, -y, 0\right]$	`[15,19]`
12	$\left[y, -y, \frac{1}{2}\right]$	`[16,18]`
13	$\left[\frac{1}{2}, -y, y\right]$	`[25,38]`
14	$\left[0, -y, \frac{1}{2}-y\right]$	`[26,33]`
15	$\left[0, y+\frac{1}{2}, y\right]$	`[27,34]`
16	$\left[y+\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$	`[28,45]`
17	$\left[-y, \frac{1}{2}-y, 0\right]$	`[29,47]`
18	$\left[y, 0, y+\frac{1}{2}\right]$	`[30,46]`
19	$\left[\frac{1}{2}-y, \frac{1}{2}, y+\frac{1}{2}\right]$	`[31,48]`
20	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}-y\right]$	`[32,37]`
21	$\left[\frac{1}{2}-y, 0, -y\right]$	`[35,44]`
22	$\left[y, \frac{1}{2}, -y\right]$	`[36,41]`
23	$\left[y+\frac{1}{2}, y, 0\right]$	`[39,43]`
24	$\left[-y, y, \frac{1}{2}\right]$	`[40,42]`

Table 11: Wyckoff site: `24k`, site symmetry: `..m'`

No.	position	mapping
1	$[x, x, z]$	`[1,36]`
2	$\left[x+\frac{1}{2}, -z, x+\frac{1}{2}\right]$	`[2,43]`
3	$\left[x+\frac{1}{2}, z+\frac{1}{2}, -x\right]$	`[3,45]`
4	$\left[z+\frac{1}{2}, x+\frac{1}{2}, -x\right]$	`[4,44]`
5	$\left[-z, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[5,46]`
6	$\left[-x, x+\frac{1}{2}, z+\frac{1}{2}\right]$	`[6,32]`
7	$\left[x+\frac{1}{2}, -x, z+\frac{1}{2}\right]$	`[7,33]`
8	$\left[x, \frac{1}{2}-x, \frac{1}{2}-z\right]$	`[8,30]`
9	$\left[\frac{1}{2}-x, x, \frac{1}{2}-z\right]$	`[9,31]`
10	$\left[\frac{1}{2}-x, \frac{1}{2}-x, z\right]$	`[10,35]`
11	$\left[x+\frac{1}{2}, x+\frac{1}{2}, -z\right]$	`[11,34]`
12	$[-x, -x, -z]$	`[12,25]`
13	$\left[-x, z+\frac{1}{2}, x+\frac{1}{2}\right]$	`[13,47]`
14	$[-x, -z, -x]$	`[14,42]`
15	$\left[z+\frac{1}{2}, -x, x+\frac{1}{2}\right]$	`[15,48]`
16	$[-z, -x, -x]$	`[16,41]`
17	$[z, x, x]$	`[17,40]`

continued ...

Table 11

No.	position	mapping
18	$[x,\ z,\ x]$	[18,38]
19	$\left[\frac{1}{2}-x,\ z,\ \frac{1}{2}-x\right]$	[19,26]
20	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ x\right]$	[20,28]
21	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ x\right]$	[21,27]
22	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	[22,29]
23	$\left[x,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[23,37]
24	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-x\right]$	[24,39]

Table 12: Wyckoff site: 48l, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$\left[x+\frac{1}{2},\ -z,\ y+\frac{1}{2}\right]$	[2]
3	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ -y\right]$	[3]
4	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ -x\right]$	[4]
5	$\left[-z,\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$	[5]
6	$\left[-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[6]
7	$\left[y+\frac{1}{2},\ -x,\ z+\frac{1}{2}\right]$	[7]
8	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[8]
9	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[9]
10	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[10]
11	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	[11]
12	$[-y,\ -x,\ -z]$	[12]
13	$\left[-x,\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	[13]
14	$[-x,\ -z,\ -y]$	[14]
15	$\left[z+\frac{1}{2},\ -y,\ x+\frac{1}{2}\right]$	[15]
16	$[-z,\ -y,\ -x]$	[16]
17	$[z,\ x,\ y]$	[17]
18	$[y,\ z,\ x]$	[18]
19	$\left[\frac{1}{2}-y,\ z,\ \frac{1}{2}-x\right]$	[19]
20	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y\right]$	[20]
21	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x\right]$	[21]
22	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	[22]
23	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[23]
24	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-y\right]$	[24]
25	$[-x,\ -y,\ -z]$	[25]
26	$\left[\frac{1}{2}-x,\ z,\ \frac{1}{2}-y\right]$	[26]
27	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ y\right]$	[27]
28	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ x\right]$	[28]
29	$\left[z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$	[29]
30	$\left[y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[30]
31	$\left[\frac{1}{2}-y,\ x,\ \frac{1}{2}-z\right]$	[31]
32	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[32]
33	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	[33]
34	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	[34]

continued ...

Table 12

No.	position	mapping
35	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z\right]$	[35]
36	$[y,\ x,\ z]$	[36]
37	$\left[x,\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	[37]
38	$[x,\ z,\ y]$	[38]
39	$\left[\frac{1}{2}-z,\ y,\ \frac{1}{2}-x\right]$	[39]
40	$[z,\ y,\ x]$	[40]
41	$[-z,\ -x,\ -y]$	[41]
42	$[-y,\ -z,\ -x]$	[42]
43	$\left[y+\frac{1}{2},\ -z,\ x+\frac{1}{2}\right]$	[43]
44	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ -y\right]$	[44]
45	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ -x\right]$	[45]
46	$\left[-z,\ x+\frac{1}{2},\ y+\frac{1}{2}\right]$	[46]
47	$\left[-y,\ z+\frac{1}{2},\ x+\frac{1}{2}\right]$	[47]
48	$\left[z+\frac{1}{2},\ -x,\ y+\frac{1}{2}\right]$	[48]